Name _____ Period _____ Date _____

7-3 Lesson Quiz: Graphing Sine and Cosine Functions

short form

Question numbers match the Savvas Lesson Quiz; questions 2 and 5 are not on this form.

1. Select **all** the correct statements.

☐ **A.** The amplitude of $y = 2\cos\left(\frac{1}{2}x\right)$ is $\frac{1}{2}$.

☐ **B.** The period of $y = 2\cos\left(\frac{1}{2}x\right)$ is π.

☐ **C.** The amplitude of $y = \frac{\pi}{2}\sin\left(\frac{\pi}{2}x\right)$ is $\frac{\pi}{2}$.

☐ **D.** The period of $y = 4\pi\sin\left(\frac{1}{2}x\right)$ is 4π.

☐ **E.** The period of $y = \cos(2\pi x)$ is 1.

☐ **F.** The amplitude of $y = 2\sin\left(\frac{1}{2}x\right)$ is 2.

Show one period calculation: _____

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3. The equation for f and the graph of g are given. How do the period and the amplitude of the functions compare?

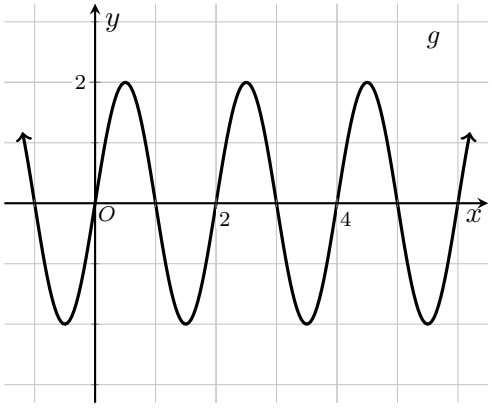
$$f(x) = 2\cos\left(\frac{\pi}{2}x\right)$$

Ⓐ The amplitudes are the same, but the period of f is half as long as the period of g.

Ⓑ The amplitudes are the same, but the period of f is twice as long as the period of g.

Ⓒ The periods are the same, but the amplitude of f is half as great as the amplitude of g.

Ⓓ The periods are the same, but the amplitude of f is twice as great as the amplitude of g.



period of f: _____

period of g: _____

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4. The graph of the function $f(x) = 5\cos(2x)$ is a reflection over the x-axis of the graph of

$$g(x) = _____\ _____(_____)\,.$$

Write one tile in each blank. | -5 | cos | $(2x)$ | sin | $(-2x)$ | 5 |

Scoring: Question 1 and Question 3 are 10 points each. Question 4 is a bonus of up to 5 points. Total out of 20.

7-3 Lesson Quiz, short form: answer key

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1. **C, D, E, F.** Amplitude $= |a|$, period $= \frac{2\pi}{|b|}$. A is false: the amplitude of $2\cos\left(\frac{1}{2}x\right)$ is 2 ($\frac{1}{2}$ is b, not a). B is false: its period is $\frac{2\pi}{1/2} = 4\pi$ (B multiplies 2π by $\frac{1}{2}$, the Try It 3 common error). C: $\left|\frac{\pi}{2}\right| = \frac{\pi}{2}$. D: $\frac{2\pi}{1/2} = 4\pi$ (the 4π in front is the amplitude and does not change the period). E: $\frac{2\pi}{2\pi} = 1$. F: $|2| = 2$. Savvas quiz item 1, DOK 1, Skills Review and Practice F90. Taught by Example 3 (Try It 3 is the same two formulas). **10 points:** 2.5 for each of C, D, E, F checked; subtract 2.5 for each of A, B checked; minimum 0.

3. **B.** f: amplitude $|2| = 2$, period $\frac{2\pi}{\pi/2} = 4$. g: midline $y = 0$, maximum 2, so amplitude 2; maxima at $x = \frac{1}{2}$ and $x = \frac{5}{2}$, so period $\frac{5}{2} - \frac{1}{2} = 2$. Same amplitude, and the period of f (4) is twice the period of g (2). Distractor A reverses the comparison; C and D say the periods are equal. Savvas quiz item 3, DOK 2, F90. Taught by Example 6 (same task with different numbers). **10 points:** 3 for the period of f with work, 3 for the period of g read off the graph, 4 for choice B.

4. **-5; cos; $(2x)$, so $g(x) = -5\cos(2x)$.** Reflecting over the x-axis replaces y with $-y$: $-\big(5\cos(2x)\big) = -5\cos(2x)$, and reflecting g back gives f. Also correct: -5; cos; $(-2x)$, because cosine is even, so $-5\cos(-2x) = -5\cos(2x)$; give it full credit. The tiles 5, cos, $(-2x)$ are a trap: $5\cos(-2x) = 5\cos(2x)$ is f itself, not its reflection. Savvas quiz item 4, DOK 1, F90. Taught by Example 3B ($y = -f(x)$ is a reflection over the x-axis) and the packet's DOK 3 Practice 23, Part E. **Bonus 5:** 3 for -5, 1 for cos, 1 for $(2x)$ or $(-2x)$.

Items left off this form (available for a retake or Form B): Savvas item 2, the average rate of change of $y = 3\sin(2x)$ on $\left[0, \frac{\pi}{4}\right]$ (answer D, $\frac{12}{\pi}$, because $\frac{3\sin(\pi/2) - 3\sin 0}{\pi/4 - 0} = \frac{3}{\pi/4}$; DOK 1, F31; it needs the quarter-period key point of the cut Example 4 and the Topic 1 rate-of-change skill); Savvas item 5, the amplitude of the tide model with high tide 5 ft and low tide 1 ft (answer 2 ft, $\frac{5-1}{2}$; DOK 2, F90; the cut Example 5, and Practice 20 is its DOK 3 cousin). Neither item has a figure; both are left off because no kept Example teaches them. Item 5 is a one-round whiteboard bridge if the teacher puts it back: amplitude $= \frac{\max - \min}{2}$, midline $= \frac{\max + \min}{2}$.

IXL replacement: SmartScore 70 or above on 9S8 and 5XN replaces this quiz with 20/20.